

Gamma Function:

$$\Gamma(n) = \int_0^{\infty} x^{n-1} e^{-x} dx$$

always from $0 \rightarrow \infty$ $n > 0$

Important formulas

$$\therefore \Gamma(n) = (n-1)!$$

$$\therefore \Gamma(n+1) = n!$$

Recursive Property

$$\therefore \Gamma(x+1) = x \Gamma(x)$$

$$\Gamma\left(\frac{1}{2}\right) = \Gamma\left(\frac{1}{2}+1\right) = \frac{1}{2} \Gamma\left(\frac{1}{2}\right)$$

$$\therefore \sqrt{\frac{1}{2}} = \sqrt{\pi}$$

$$= \frac{1}{2} \sqrt{\pi}$$

$$\textcircled{a}) \Gamma_5$$

$$= (5-1)!$$

$$= 4!$$

$$\Gamma_{4+1}$$

$$= 4!$$

$$\textcircled{b}) \text{ Evaluate}$$

$$\int_0^{\infty} x^{\frac{1}{2}} e^{-\sqrt{x}} dx$$

$$\therefore \Gamma(n) = \int_0^{\infty} x^{n-1} e^{-x} dx$$

$$u = \sqrt{x}$$

$$x = u^2$$

$$u = \sqrt{x}$$

$$u = x^{\frac{1}{2}}$$

$$du = \frac{1}{2\sqrt{x}} dx$$

$$dx = 2\sqrt{x} du$$

$$\int_0^{\infty} (u^2)^{\frac{1}{2}} e^{-u} 2\sqrt{x} du$$

$$\int_0^{\infty} u^{\frac{1}{2}} \cdot e^{-u} \cdot u du$$

$$\int_0^{\infty} u^{\frac{3}{2}} \cdot e^{-u} du$$

$$\int_0^{\infty} u^{\frac{3}{2}-1} \cdot e^{-u} du$$

$$= \Gamma\left(\frac{3}{2}\right)$$

$$= \Gamma\left(\frac{3}{2}\right)$$

$$= \sqrt{\frac{3}{2}+1}$$

$$= \frac{3}{2} \sqrt{\frac{3}{2}}$$

$$= \frac{3}{2} \times \frac{1}{2} \sqrt{\frac{1}{2}+1}$$

$$= \frac{1}{4} \Gamma\left(\frac{1}{2}\right) = 2^{\frac{1}{2}} \frac{3}{4} \sqrt{\pi}$$

$$e) \int_0^{\infty} \sqrt{x} e^{-3\sqrt{x}} dx \quad \therefore \Gamma(n) = \int_0^{\infty} x^{n-1} e^{-x} dx$$

$$\begin{array}{l|l} u = 3\sqrt{x} & \sqrt{x} = \frac{u}{3} \\ u = 3x^{\frac{1}{2}} & \\ du = 3 \cdot \frac{1}{2} \frac{1}{\sqrt{x}} dx & \int_0^{\infty} \frac{u}{3} e^{-u} \cdot \frac{2}{3} x \frac{u}{3} du \\ du = \frac{3}{2\sqrt{x}} dx & \frac{2}{27} \int_0^{\infty} u^2 \cdot e^{-u} du \\ dx = \frac{2\sqrt{x}}{3} du & \frac{2}{27} \int_0^{\infty} u^{2-1} \cdot e^{-u} du \\ & \frac{2}{27} \Gamma_3 \\ & \frac{2}{27} (3-1)! \\ & = \frac{2}{27} \end{array}$$

$$e) \int_0^{\infty} \sqrt{x} e^{-x} dx \quad \therefore \Gamma(n) = \int_0^{\infty} x^{n-1} e^{-x} dx$$

$$\begin{array}{l|l} u = x & \int_0^{\infty} u^{\frac{1}{2}} \cdot e^{-u} du \\ du = 1 dx & = \Gamma_{\frac{3}{2}} \\ & = \frac{1}{2} \Gamma_{\frac{1}{2}+1} \\ & = \frac{1}{2} \sqrt{\pi} \end{array}$$

$$e) \int_0^{\infty} x^2 \cdot e^{-5x} dx \quad \therefore \Gamma(n) = \int_0^{\infty} x^{n-1} e^{-x} dx$$

$$\begin{array}{l|l} u = 5x & x = \frac{u}{5} \\ du = 5 dx & \int_0^{\infty} \frac{u^2}{25} \cdot e^{-u} \cdot \frac{u}{5} du \\ du = \frac{1}{5} du & = \frac{1}{125} \Gamma_3 \\ & = \frac{2}{125} \end{array}$$

$$e) \int_0^{\infty} x^5 \cdot e^{-x} dx$$

$$\int_0^{\infty} x^{6-1} \cdot e^{-x} dx$$

$$\Gamma_6 =$$

$$= 5!$$

o) Prove $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$

$$\int_0^{\pi/2} \sin^p x \cdot \cos^q x \, dx = \frac{\Gamma\left(p + \frac{1}{2}\right) \cdot \Gamma\left(q + \frac{1}{2}\right)}{2 \cdot \Gamma\left(\frac{p+q+2}{2}\right)}$$

$$p = q = 0$$

$$\int_0^{\pi/2} \sin^0 x \cdot \cos^0 x \, dx = \frac{\Gamma\left(\frac{1}{2}\right) \cdot \Gamma\left(\frac{1}{2}\right)}{2 \cdot \Gamma\left(\frac{2}{2}\right)} = \frac{\left(\Gamma\left(\frac{1}{2}\right)\right)^2}{2}$$

$$\int_0^{\pi/2} 1 \cdot dx = \frac{1}{2} \left(\Gamma\left(\frac{1}{2}\right)\right)^2$$

$$1 \cdot \pi \cdot \frac{1}{2} = \frac{1}{2} \left(\Gamma\left(\frac{1}{2}\right)\right)^2$$

$$\frac{\pi}{2} = \frac{1}{2} \left(\Gamma\left(\frac{1}{2}\right)\right)^2$$

$$\sqrt{\pi} = \Gamma\left(\frac{1}{2}\right)$$

o) $\int_0^{\infty} x^n \cdot e^{-x^2} \, dx$

$$v = x^2 \quad x = \sqrt{v}$$

$$dv = 2x \, dx \quad x = v^{1/2}$$

$$\frac{1}{2} dv = dx$$

$$\therefore \Gamma(n) = \int_0^{\infty} x^{n-1} \cdot e^{-x} \, dx$$

$$\int_0^{\infty} (v^{1/2})^1 \cdot e^{-v} \cdot \frac{1}{2(v^{1/2})} \, dv$$

$$\frac{1}{2} \int_0^{\infty} \frac{v^2}{v^{1/2}} \cdot e^{-v} \, dv$$

$$\frac{1}{2} \int_0^{\infty} v^{\frac{2}{2}-1} \cdot e^{-v} \, dv$$

$$\frac{1}{2} \Gamma\left(\frac{2}{2}\right)$$

$$\frac{1}{2} \Gamma\left(\frac{1}{2} + 1\right)$$

$$\frac{1}{2} \times \frac{2}{2} \Gamma\left(\frac{1}{2} + 1\right)$$

$$\frac{1}{2} \times \frac{2}{2} \times \frac{1}{2} \Gamma\left(\frac{1}{2}\right)$$

$$= \frac{3}{8} \sqrt{\pi}$$

8) Prove that $\Gamma(1) = 1$

$$\int_0^{\infty} x^{n-1} \cdot e^{-x} dx$$

$$\therefore \int e^{-x} = -e^{-x} = -\frac{1}{e^x}$$

$$\int_0^{\infty} x^{1-1} \cdot e^{-x} dx$$

$$\int_0^{\infty} x^0 \cdot e^{-x} dx$$

$$\int_0^{\infty} e^{-x} dx$$

$$\left| -\frac{1}{e^x} \right|_0^{\infty}$$

$$= \left| -\frac{1}{e^{\infty}} + \frac{1}{e^0} \right| \quad \therefore e^{\infty} = \infty, e^0 = 1$$

$$= \left| -\frac{1}{\infty} + \frac{1}{1} \right|$$

$$= | -0 + 1 |$$

$$= 1$$

